Analysis of Construction Safety Trends, Ergonomic Risk and Hazard Mitigation

A Comprehensive Report on Fatal and Non-Fatal Injury Trends, Ergonomic Hazard Identification, and Evidence-Based Interventions in the Construction Industry

Prepared by

Konrad Yee and Keion Savary

December 3rd, 2025

Section 1.1 - Analysis of Construction Safety Trends and Hazard Mitigation

Over the past decade, particularly in the USA, the number of fatalities has been gradually increasing from 781 in 2011 to 1092 in 2022 (Figure 1). However, the fatality rate has hovered around 9.5 per 100,000 full-time equivalents (FTE) (Figure 1). Overall, this shows that there has been a long pattern of fluctuations between a high number of fatalities and no immediate action to attempt to reduce them. In addition to the fatalities, the trend in non-fatal injuries has seen a small decrease to 84.4 FTE from 109.1 FTE in 2015-16 over the past decade [13]. Generally speaking, the non-fatal injuries have been stable and are bound to happen when working in a hectic environment, working with machinery and lifting heavy loads over a long period of time.

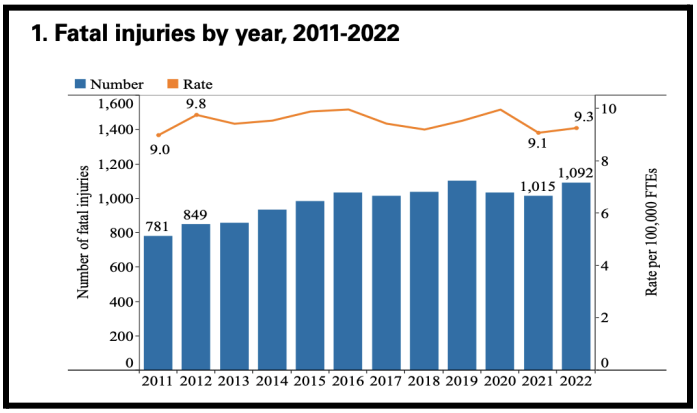


Figure 1. Diagram of fatal injuries by year

Year over year no consistent improvement in safety by the numbers of fatalities. The trend for fatal counts and rates in the construction sector has stayed consistent, hovering around 9.5 deaths per 100,000 people for 2013-2023. This pattern indicates no clear, or at least no impactful year over year improvement in fatal outcomes. There have been modest improvements to the non-fatal outcomes in some sectors, but the sum of non-fatal construction incidence remains high (2.3/100 people). Particular trades such as framing, roofing, and powerlines still show elevated rates.

The construction safety record is significantly higher compared to other major industries like manufacturing, retail, and agriculture. According to the U.S Bureau of Labour Statistics, construction had 1075 fatal injuries, while agriculture had 448, manufacturing had 391, and retail had 306 in 2023 [11]. One of the main causes of the high amount of fatalities has been due to falls, slips and trips, accounting for 39% of fatalities [12]. Many of these fatalities could be due to a lack of safety training, not wearing proper personal protective equipment or just general accidents.

However, advancements in technology are playing a big role in keeping construction workers safe on the job across all worksites. Advancements in video analysis tools can help mitigate the risk of fatal injuries. They analyze the current workspace and can identify any changes in safety protocols, alerting workers immediately. Similarly, drones can capture and monitor the workspace to identify any potential hazards, and they can also track bad habits and patterns within the workspace. Finally, building information modelling (BIM) gives you a 3D

visualization of the project, which allows you to analyze different scenarios to help reduce potential hazards and any complications present [10].

Section 1.2 - Ergonomic Risk Analysis of High-Strain Construction Tasks

The sole proprietor for a high risk of musculoskeletal strain, primarily involving the act of forceful exertion, acting in awkward postures and repetitive motions, especially heavy lifting. Alongside constant poor ergonomics and postures, construction tasks account for common back injuries and fatigue affecting muscles, joints and posture. Building on the findings from Task 1, two tasks that present high ergonomic strain are lifting materials from the ground and laying floor tiles.

Lifting materials from ground level is a common task when working in construction. This task requires you to bend your hips, flex your spine and generate enough force from your quadriceps and hamstrings to lift the material (Figure 2). The maximum threshold that a person could lift is approximately 51 pounds [19]; however, in a construction workspace, workers are seen lifting heavy loads for prolonged periods of time, which increases the chances of developing work-related musculoskeletal disorders (WMSDs). Ergonomically, this task is hazardous due to the repetition at heavy weight, the required force to lift and the awkward posture of bending forward.

Figure 2. Diagram of how to properly lift a box

Tile installation is another high-risk ergonomic task. Workers tend to stay in a low, bent-over, kneeling position for extended periods of time, which alters their natural posture and can put immense pressure on their knees and neighbouring joints. Research shows that extended periods of kneeling can cause a higher risk of developing patellofemoral cartilage damage, resulting in patellofemoral osteoarthritis [14]. Ergonomically, this task is hazardous due to prolonged periods of awkward and static postures and repetition.

Across both tasks, there are several gaps and challenges in how ergonomic risks are managed. A major gap is the insufficient access to ergonomic training. A lot of workers are not taught the proper way to do tasks to limit the potential WMSDs that they can later develop. One challenge with implementing ergonomic training is ensuring that the training is engaging and interactive for the workers. If the training doesn’t appeal to them or doesn't keep them engaged, they won't learn the proper way to do tasks [15]. Another gap is cost allocation. The ergonomic programs require funding, which the private industries may not provide. One challenge with the ergonomic

programs is that companies will not have time to implement the training due to their busy schedules with many different projects that need to be completed by a certain date.

Section 2. Task Description, Justification, and Data-Collection Workflow

A common physically demanding construction task like tile installation exposes workers to awkward postures, high hand-force exertions, prolonged kneeling and prolonged spine flex positions. These postures increase the likelihood of work-related musculoskeletal disorders (WMSDs) by putting biomechanical loads on the spine, knees, shoulders, and upper limbs (Figure 3). This section (1) justifies selecting the analysis task of tile installation, (2) outlines the workflow to collect and calculate joint-angle data, and (3) identifies key limitations that may affect the accuracy of the motion-capture data used for the REBA assessment.

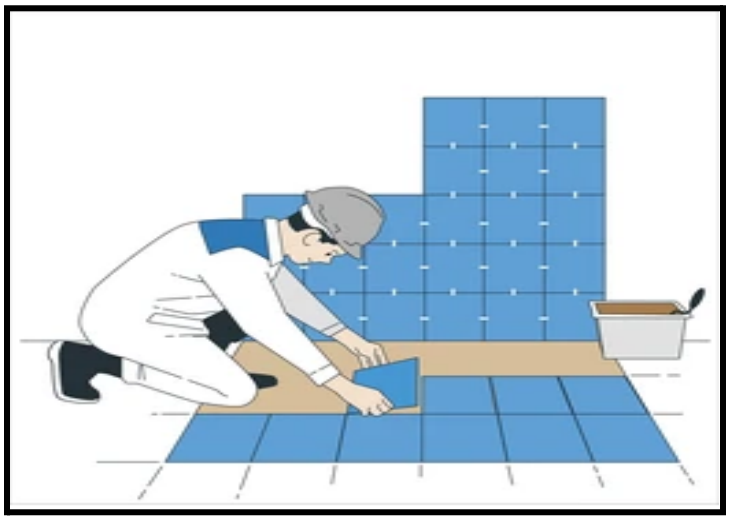


Figure 3. Diagram of tile installation

2.1 Task Justification

Tile installation with high ergonomic risk was a good selection as it is well-documented. Industry injury data shows that tile workers exhibit WMSD rates of approximately 150 cases per 10000, nearly double the sector average [7]. Furthermore, ergonomic research found that 91% of tile installers meet the caution threshold for high exposures in multiple body parts [17].

Repetitive kneeling, squatting, forward trunk flexion, and the use of force when spreading or hammering mortar are all necessary for tile installation. These factors align with ergonomic risk patterns identified earlier in the report: long-duration flexion, high repetition, and high-force tool use. The task is a perfect case study for analyzing motion-capture joint-angle data and REBA scoring because it naturally results in significant deviation from neutral posture.

2.2 Evidence of Ergonomic Strain During Tile Installation

Research shows trunk flexion exceeding 20-30° while holding tiles statically substantially increases spinal loading [1]. Workers often maintain unsafe trunk flexion exceeding 45° during tile setting. Furthermore, it is shown that there is overextension in tool use exceeding 90° flexion in frequent kneeling and squatting, sustained postures near end-range knee flexion, and shoulder

flexion/abduction [2]. Further proving that tile installation causes joint angles with ranges that significantly elevate WMSD risk.
For comparison, the joint angles we measured for action frames align with expected anatomical ranges reported for angle joint research on tile laying, for example, kneeling postures commonly produce knee flexion angles of 70-100°, consistent with our calculated flexion of 72° in kneeling frames(frame 161) [3]. This agreement validates our angle data while acknowledging that markerless motion capture is inherently approximate.

2.3 Data-Collection Workflow

2.3.1 Video Recording and Motion-Capture Setup

In a controlled workspace to replicate actual working conditions, imitation of tile installation motions was recorded. Two video cameras were positioned and synced around the subject to minimize occlusion and provide full-body visibility. A subject imitated a typical tile installation sequence, including reaching, kneeling, placing tiles, applying mortar, and tapping tiles into place. To get enough information for the software, the recording lasted about ten seconds.

2.3.2 Export and Blender Reconstruction

The video footage was processed using a markerless motion-capture system (Rokoko), which estimated 3D joint centres for each frame. After importing these joint centres into Blender, a Python script from the TP4 Toolkit on Quercus exported a CSV file with 3D coordinates for every major joint in every frame. This step produced the raw spatial data needed for subsequent joint-angle calculations.

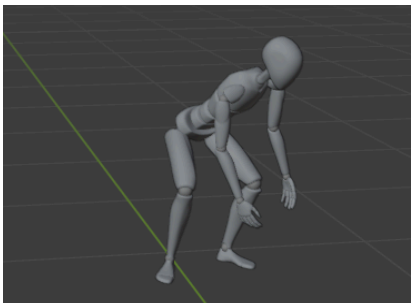
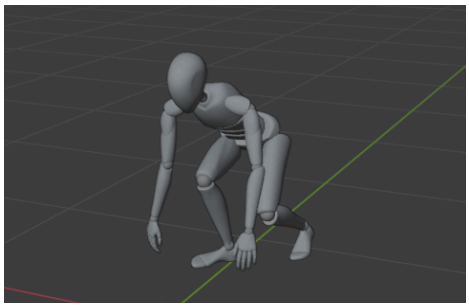
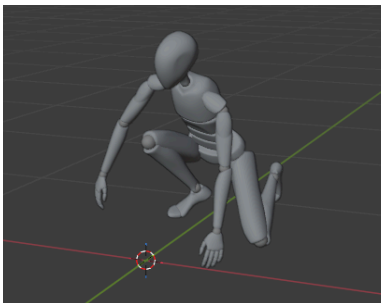
2.3.3 Joint-Angle Calculation

Joint-angle calculations were completed using a Jupyter Notebook. Using vector dot-products between neighbouring anatomical segments (such as thigh–shank and trunk–pelvis), the script calculated angles. The output was a second CSV containing calculated joint angles for each frame. Later, representative postures and REBA scoring were determined using these angles.

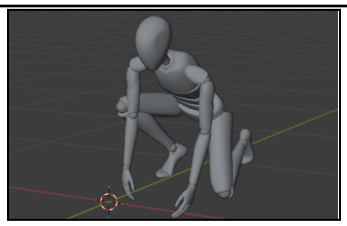
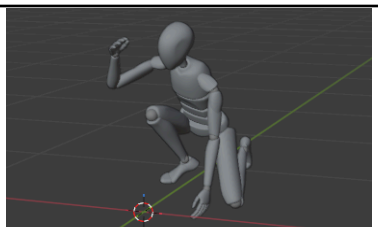
2.3.4 Representative Posture Identification

Five key frames were selected to reflect critical ergonomic positions observed during tile installation:

Frames 1-3:

		
1. Standing reach to pick up a tile	2. Forward-flexing transition into kneeling	3. Kneeling while applying mortar

Frames 4-5:

	
4. Squatting to place a tile on the floor	5. Extended arm posture during hammering

2.4 Limitations of the Motion-Capture and Processing Method

While video-based markerless MoCap offers naturalistic movement and minimal interference, it has limitations that influence the accuracy of joint-angle estimation.

2.4.1 Markerless Mocap Accuracy

Markerless pose estimation using the camera setup, depending on the joint and motion, can produce joint angle errors ranging from 3-7°[4]. The reduced precision of markerless systems particularly affects: Hip angles, Ankle and wrist dorsiflexion/eversion and Trunk flexion (depth perception errors). Additionally, with inconsistencies in the MoCap not sensing the floor properly, wrist motions were especially affected; the MoCap often misestimates subtle rotations, leading to minor inconsistencies in distal-joint angles. Overall, our data supported typical strains and angles of joints in the common practice of tile laying.

2.4.2 Limitations of Replicating Tile Laying

The joint angle accuracy is reduced through imitation of tasks. Since replicating tile laying has the absence of the tile weight and tool grip, the joint angles and overall stress naturally produce smaller values, particularly in the lumbar spine and wrists. Tile setting in practice requires greater loads, which causes difficulty in reproduction for simulation, i.e. lifting several kilogram tiles, applying downward forces with hands and stabilizing the trunk under load. Regardless of limitations, the reconstructed motion produced accurate data for essential postures (deep knee flexion, forward trunk lean, and floor-level reaching) for the REBA table analysis. The data provides a reasonable approximation of the task for ergonomic analysis when interpreted with these constraints in mind.

2.4.3 Impact of Limitations on REBA Analysis

REBA categorizes posture into broad angle bins (e.g., trunk flexion 20–60°), small angular fluctuations (<5°) do not significantly alter scoring. However, larger systematic errors (e.g., underestimating trunk flexion from 50° to 35°) could lead to misclassification of risk levels. Therefore, wherever possible, observational notes and posture screenshots will complement numerical angles to ensure accurate interpretation.

2.5 Summary

Tile installation is a high-risk musculoskeletal task strongly supported by quantitative injury data and ergonomic literature. The multi-camera markerless MoCap workflow provides a feasible method for capturing realistic tile-installation movements and generating joint-angle data for analysis. Despite limitations inherent in markerless systems and Blender reconstruction, the calculated angles fall within expected anatomical ranges for tile-laying postures, supporting their use for REBA scoring.

Section 3

Section 3.1

The joint angle data were collected by using MoCap. This recording was then further analyzed for key joints like arms, legs, neck and spine. Figure 4 represents the joint angles throughout the task, and at frames 45 (natural) and 161 ("worst") were further analyzed using REBA to identify the ergonomic risks.

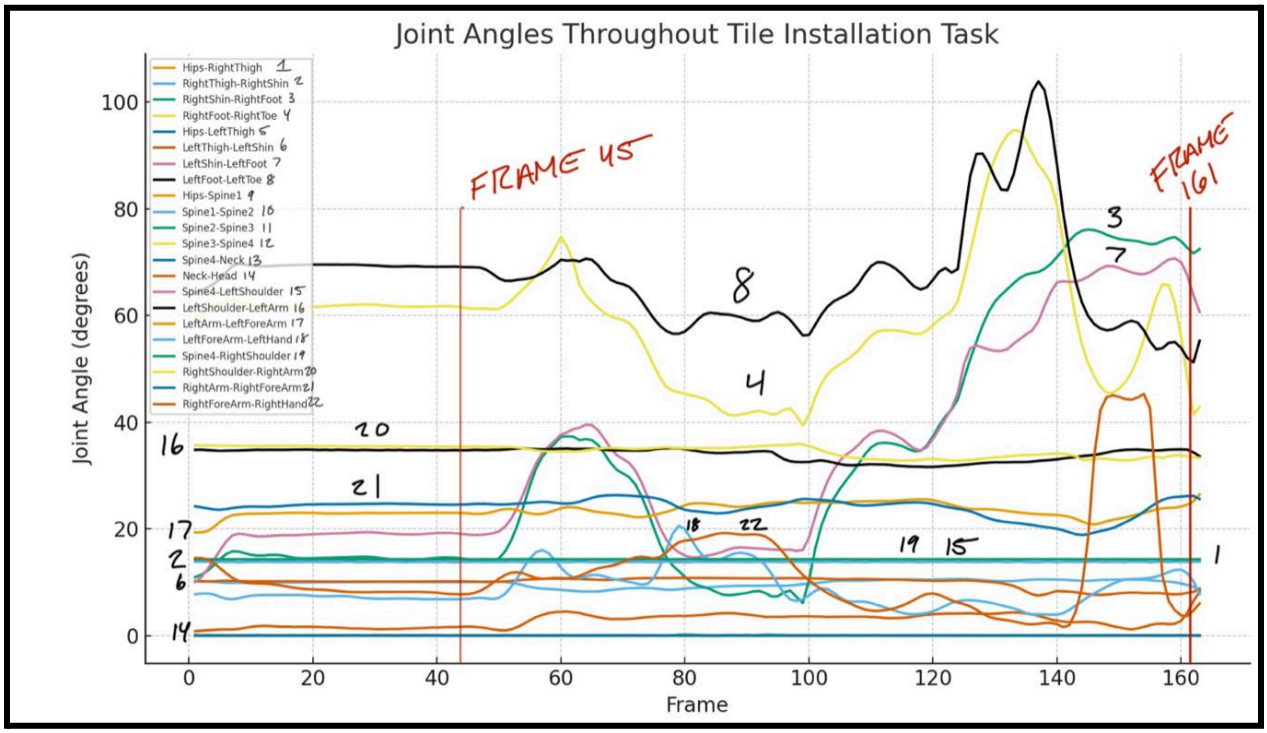


Figure 4. Plot of the Joint angles throughout the tile installation task

Note: Hips-RightThigh, Hips-LeftThigh, Hips-Spine1, Spine2-Spine3, Spine3-Spine4 and Spine4-Neck all are 0°

The REBA score was first obtained by comparing the calculated angles in the Excel file to the REBA chart. The angles between the natural and “worst” position were compared to the given figures in the chart and were given a score based on which range of degrees they fell in. Additional scores were given if there was any twist, side bending, loads and coupling. As shown in Figure 5 and Table 1, the natural position received a score of 1, which is deemed the natural position to have negligible risk.

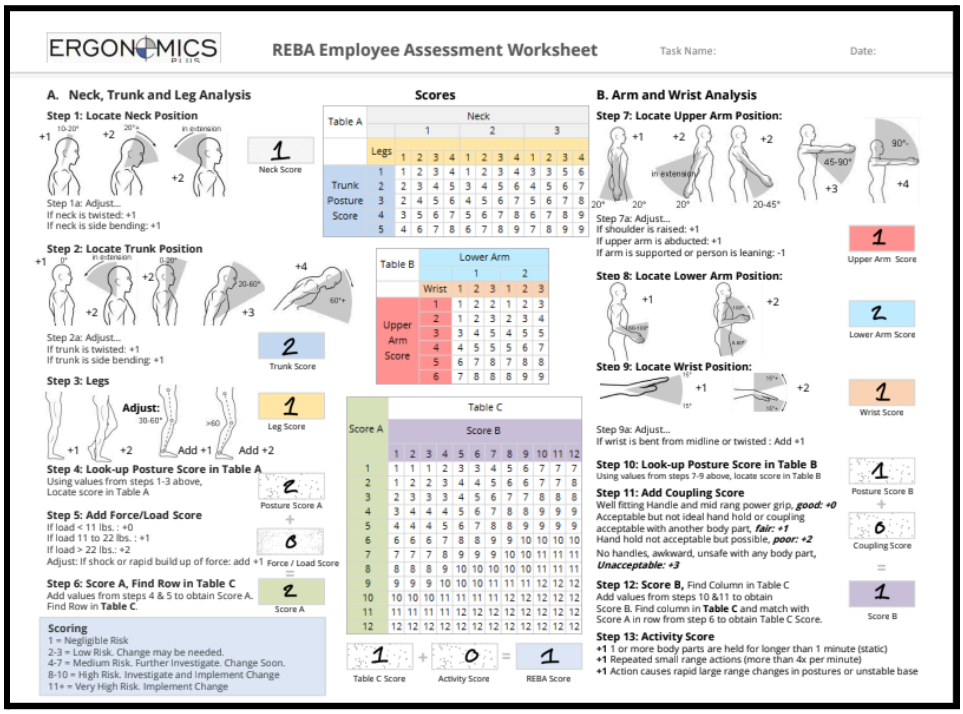


Figure 5. REBA score for natural position

As shown in Figure 6 and Table 1, the “worst” position received a score of 11, which resulted in the task having a very high risk and a need to implement a change in the way the task is performed.

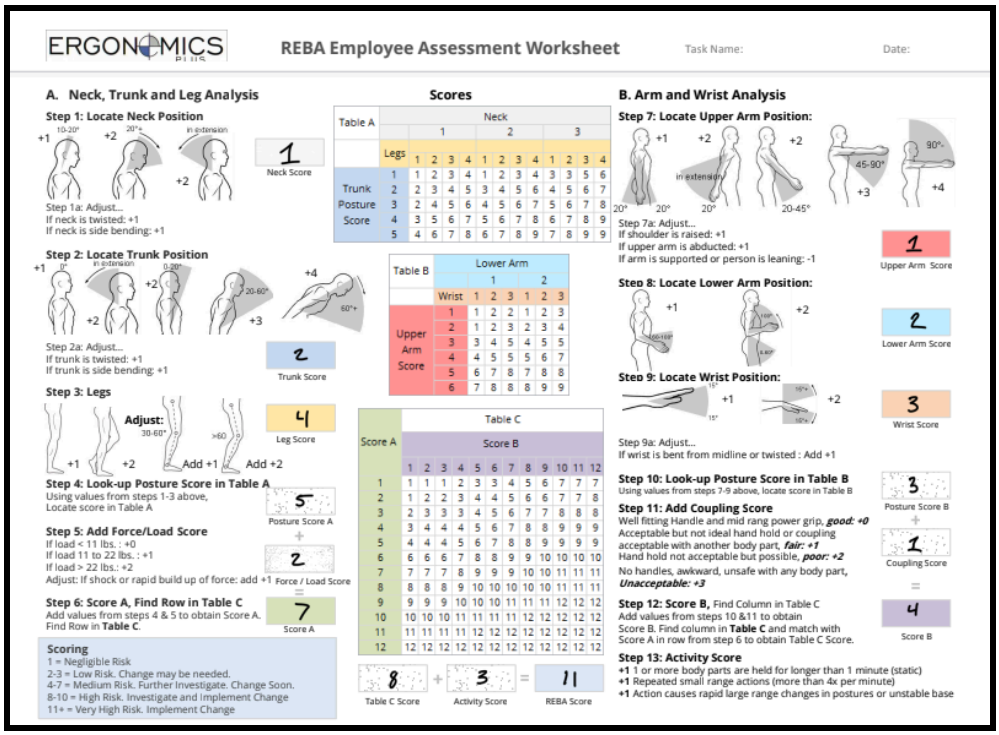


Figure 6. REBA score for "worst" position

As shown above, the natural posture during the tile installation task showed low joint angles across the neck, legs, upper arm and wrists. As a result of these low scores, the overall REBA score was relatively low (1), which is reasonable since the body is at rest. The high ergonomic risk posture was significantly higher than the natural position. The main leaders of the REBA score were the legs, force, wrists and activity. Having a joint angle flexion of more than 60°+ in the knees significantly increases the overall score and relates to the kneeling posture that the workers are constantly in. Because tiles can vary in weight, the angle of flexure for the wrists can also vary. The heavier the weight, the higher the angle and REBA score. By taking the average weight of a 24x24 inch tile (25 pounds), the angle of flexion is more than 15° and increases the score by 2 (Table 1).

Table 1: Summary of REBA scores for neutral and high-strain postures.

Body Segment	Neutral Posture	High-Strain Posture
Neck	1	1
Trunk	2	2
Legs	1	4
Posture score	2	5
Force	0	2

Score A	2	7
Upper Arm	1	2
Lower Arm	1	1
Hand	1	3
Posture Score	1	3
Coupling	0	1
Score B	1	3
Table C	1	8
Activity	0	3
Final REBA Score	1	11

Note: the assumed weight for the force was a 24x24 inch tile that weighs 25 lbs [18]

As much as the high ergonomic task is expected to be higher than the natural position, it was unexpected that the score for part A was 7. Although flexion in the knee is a clear cause of WSMDs in this task, the added weight contributes to the overall risk by adding additional pressure to the knee. Similarly, the combined effect of multiple joints being in flexion simultaneously produces a higher risk of strain and injuries than a single-jointed task. This demonstrates how REBA takes into consideration the number of joints moved, not just the extreme angles that are in flexion.

Consequently, repeating this high-risk task will heavily take a toll on your posture and health over the long term. Some complications may include:

1. Work-Related Musculoskeletal Disorders (WMSDs)
    a. WMSDs can cause excruciating pain in muscles, tendons, ligaments, nerves and discs
    b. WMSDs can also cause patellofemoral cartilage damage, resulting in patellofemoral osteoarthritis, as mentioned before
2. Chronic Fatigue
    a. When performing these high-ergonomic tasks for long periods of time, fatigue can settle in, which will ultimately increase the risk of muscle strains and injuries

b. Fatigue can lead to fatality when not working in proper conditions, and due to errors while working on the job

Furthermore, using REBA scoring logic, typical technique errors or task demands would increase the REBA score significantly. If a worker were to bend further than the trunk angle threshold, additional points would be added for unsupported and unstable posture. In the same frame of mind, if a worker were to bend their legs past a reasonable margin that's higher than 60°, it would alter the REBA score, ultimately increasing the overall score. Even lifting a moderately heavier load can increase the total REBA because the load also amplifies all the other risk factors. Changes to these scores can mean that the risk can change from a high risk to a very high risk, meaning that an urgent implementation would need to happen.

3.2 Suggestions for Improvement

The REBA analysis defined main contributors to high ergonomic risk during the construction task as follows: identified trunk flexion, asymmetric leg stance, and extended arm positions. These postures under prolonged exposure increase the likelihood of musculoskeletal injuries in the lower back, knees, and upper limbs. Two realistic proposals, one technology-based and one non-technology-based, are suggested to improve ergonomics while staying practicable for construction environments.

3.2.1 Technology-Based Recommendation: Adjustable Lift / Super Lift

Extended trunk flexion and arm reach were key drivers of high REBA scores. Introducing an adjustable lift or super lift can raise materials to waist height or allow for ease of placement, reduce trunk flexion from approximately 60–70° to 20–30°, and keep loads close to the body to lower arm extension and distribute stress evenly (Figure 7).

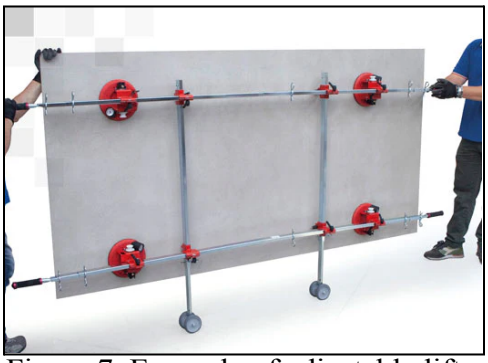


Figure 7. Example of adjustable lift

Implementation requires minimal training and can be standardized for repetitive or heavy lifting tasks. This intervention reduces strain on the trunk and shoulders, lowers injury risk,and maintains productivity.

3.2.2 Non-Technology Recommendation: PPE and Worker Training on Lifting and Posture

Ergonomic safety is heavily influenced by human factors, even when it comes to equipment. Posture abnormalities include forward bending, excessive knee flexion, and asymmetric stance significantly enhance risk, according to the REBA data. A practical, non-technological approach is focused worker training on safe lifting techniques and postural awareness.

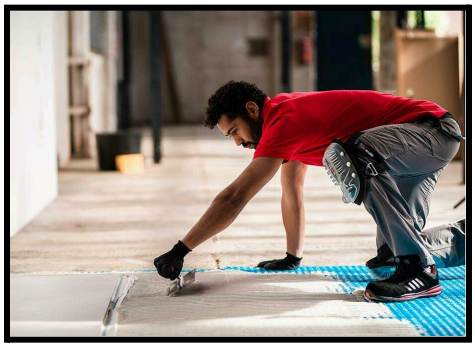


Figure 8. Illustration of proper lifting posture with PPE.

Key elements include: keeping the load close to the body to reduce arm extension and shoulder stress; adopting wider stances to improve lower limb stability and minimize asymmetric leg strain; bending at the knees rather than the trunk to limit forward flexion; alternating tasks or incorporating micro-breaks to prevent repetitive strain in the spine and upper limbs.

Workers can maintain near-neutral postures through attention to good technique, hence lowering REBA scores for the trunk, legs, and arms. This intervention is cost-effective and quickly implementable, needing only short training sessions and appropriate PPE . Over weeks and months, these modifications can considerably reduce musculoskeletal strain, boost comfort, and sustain long-term occupational health.

3.2.3 Integration and Practical Impact

A combination of both interventions provides optimal risk reduction. Utilizing the mechanical lift mitigates trunk and arm strain, while training and adding PPE ensures proper posture and task execution. Together, they create a safer, more ergonomic work environment, reduce musculoskeletal injuries, and support sustained productivity across construction tasks.

References

[1] M. P. de Looze, L. K. Kingma, and J. H. van Dieën, "Flexion–relaxation during lifting: Relations with low-back pain and motor control," *Eur. J. Appl. Physiol.*, 1993. [Online]. Available: https://pubmed.ncbi.nlm.nih.gov/8484154/. Accessed: Dec. 3, 2025.

[2] Y. Bao, B. E. Howard, and W. S. Marras, "The biomechanical demands of manual tile installation," *J. Biomech.*, vol. 46, no. 10, 2013. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0003687013002275. Accessed: Dec. 3, 2025.

[3] L. Punnett and D. H. Wegman, "Work-related musculoskeletal disorders: The epidemiologic evidence and the debate," *J. Electromyogr. Kinesiol.*, 2004. [Online]. Available: https://pmc.ncbi.nlm.nih.gov/articles/PMC4040370/. Accessed: Dec. 3, 2025.

[4] J. J. Dowell et al., "Biomechanics of occupational lifting tasks," *J. Biomech.*, 2022. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0021929022003797. Accessed: Dec. 3, 2025.

[5] CPWR, "Data Bulletin—July 2024," 2024. [Online]. Available: https://www.cpwr.com/wp-content/uploads/DataBulletin-July2024.pdf. Accessed: Dec. 3, 2025.

[6] IHSA, "Construction Death Review: Annual Report of Fatalities – Falls From Height 2025," 2025. [Online]. Available: https://www.ihsa.ca/pdfs/topics_hazards/construction-death-review-annual-report-of-fatalities-falls-from-height-2025.pdf. Accessed: Dec. 3, 2025.

[7] U.S. Bureau of Labor Statistics, "Injury and illness rates by industry," 2025. [Online]. Available: https://www.bls.gov/web/osh/table-1-industry-rates-national.htm. Accessed: Dec. 3, 2025.

[8] U.S. Bureau of Labor Statistics, "Census of Fatal Occupational Injuries Summary," 2025. [Online]. Available: https://www.bls.gov/news.release/pdf/cfoi.pdf. Accessed: Dec. 3, 2025.

[9] S. Wang et al., "Wearable IMU-based joint analysis for ergonomic assessments," *Sensors*, vol. 25, no. 16, 2025. [Online]. Available: https://www.mdpi.com/1424-8220/25/16/5201. Accessed: Dec. 3, 2025.

[10] Procore, "Construction safety technology," 2025. [Online]. Available: https://www.procore.com/library/construction-safety-technology. Accessed: Dec. 3, 2025.

[11] U.S. Bureau of Labor Statistics, "Fatal work injuries by industry," 2025. [Online]. Available:

https://www.bls.gov/charts/census-of-fatal-occupational-injuries/number-and-rate-of-fatal-work-injuries-by-industry.htm. Accessed: Dec. 3, 2025.

[12] BD+C Network, "Construction workplace deaths in 2023 rose to highest levels since 2011," 2024. [Online]. Available: https://www.bdcnetwork.com/home/news/55253385/construction-workplace-deaths-in-2023-rose-to-highest-levels-since-2011. Accessed: Dec. 3, 2025.

[13] CPWR, "Fatal and nonfatal injuries in construction: 7th Edition Chart Book," 2025. [Online]. Available: https://www.cpwr.com/research/data-center/the-construction-chart-book/interactive-7th/injuries-illnesses-health/fatal-and-nonfatal-injuries/. Accessed: Dec. 3, 2025.

[14] A. Vieira et al., "Ergonomic risk and musculoskeletal disorders," *Ergonomics*, 2020. [Online]. Available: https://pubmed.ncbi.nlm.nih.gov/33009589/. Accessed: Dec. 3, 2025.

[15] SBN Software, "Challenges of implementing an ergonomics program," 2025. [Online]. Available: https://sbnsoftware.com/blog/what-are-the-challenges-of-implementing-an-ergonomics-program/. Accessed: Dec. 3, 2025.

[16] IHSA, "W318 – Manual Material Handling Guide," 2025. [Online]. Available: https://www.ihsa.ca/PDFs/Products/Id/W318.pdf. Accessed: Dec. 3, 2025.

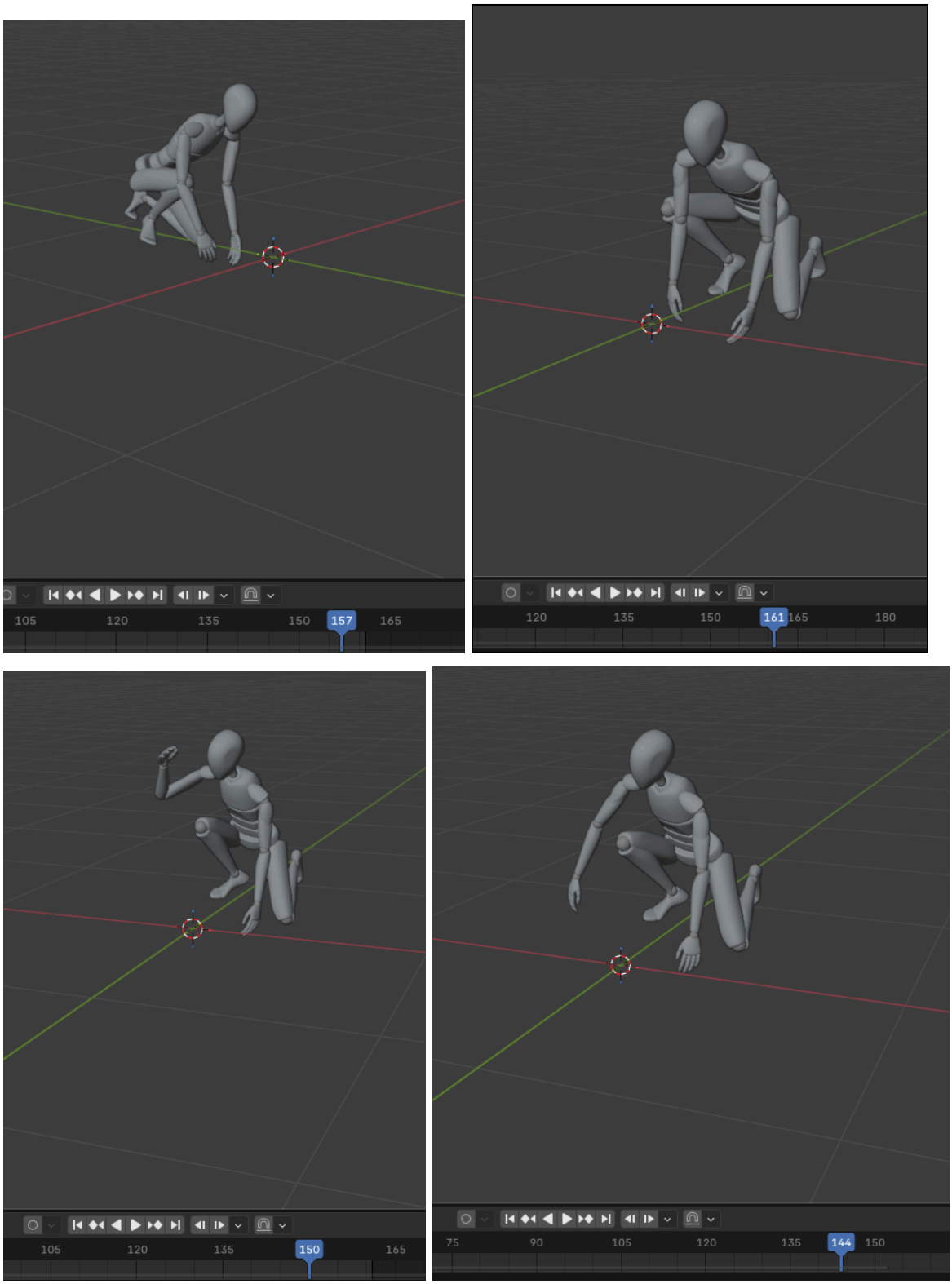
[17] E. Waters et al., "Revised NIOSH lifting equation," *Ergonomics*, 1994. [Online]. Available: https://pmc.ncbi.nlm.nih.gov/articles/PMC3999260/. Accessed: Dec. 3, 2025.

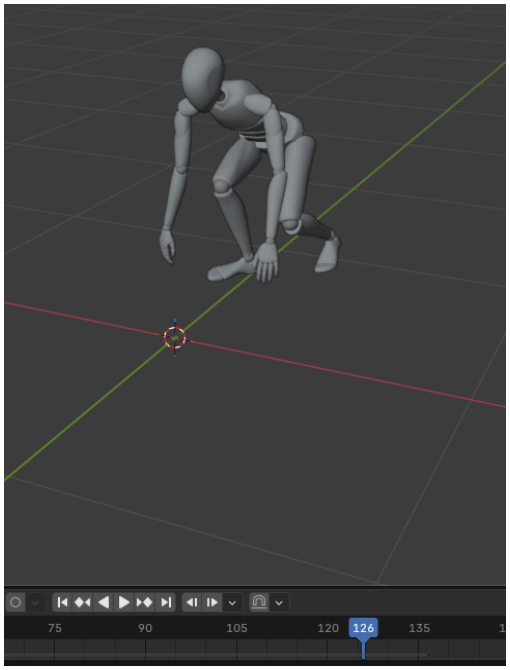
[18] LX Hausys, "How much does porcelain tile weigh?" 2025. [Online]. Available: https://www.lxhausys.com/us/blog/how-much-does-porcelain-tile-weigh/. Accessed: Dec. 3, 2025.

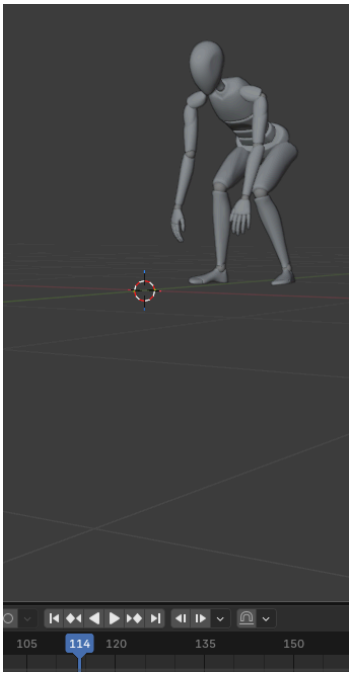
[19] CCOHS, "Assessing ergonomics in the workplace," 2025. [Online]. Available: https://www.ccohs.ca/oshanswers/ergonomics/niosh/assessing.html. Accessed: Dec. 3, 2025.

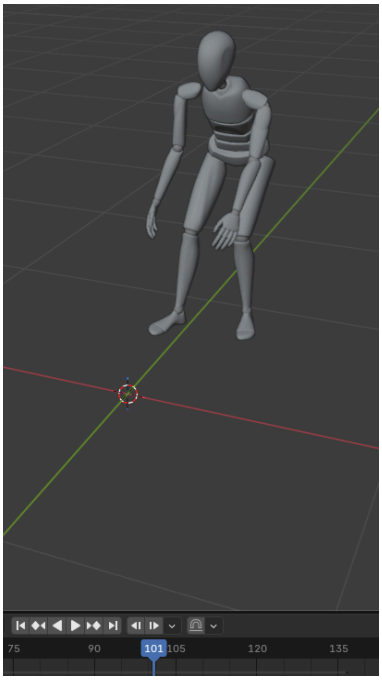
Appendix

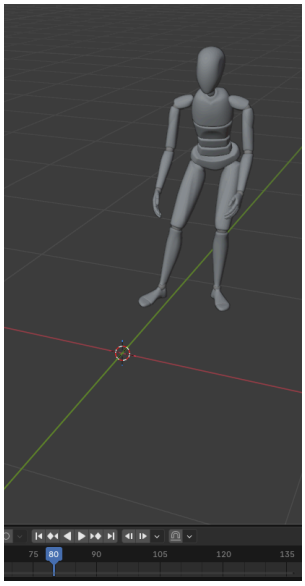
Blender Images for Laying Tiles with Matching Frames at the Bottom of the Image

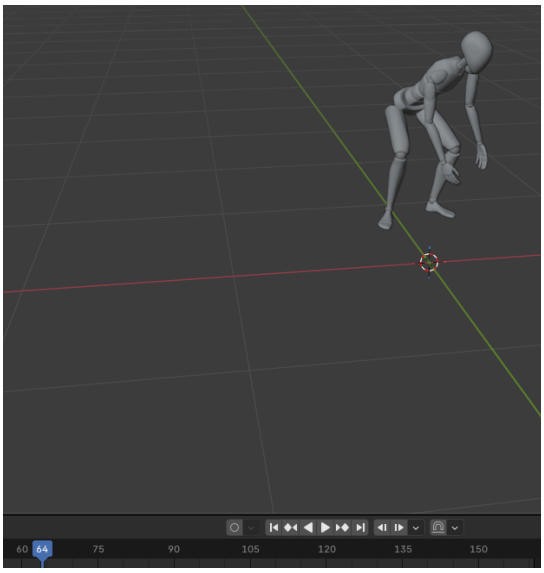


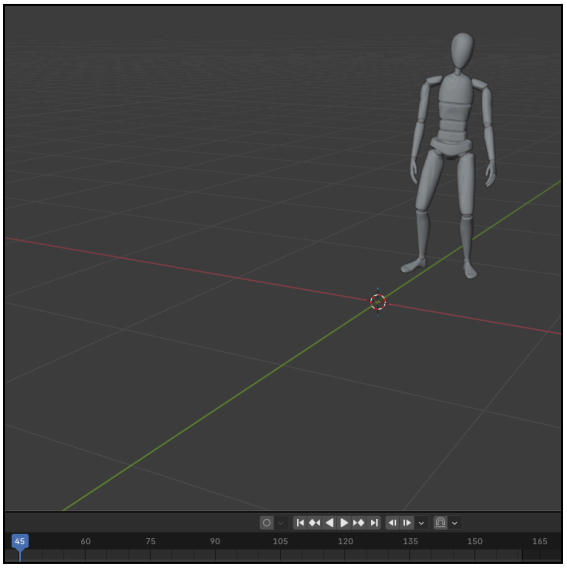

75
90
105
120
126
135


105
114
120
135
150


75
90
101
105
120
135


75
80
90
105
120
135


60
64
75
90
105
120
135
150


45
60
75
90
105
120
135
150
165